

E10235625_2

The equation $x^4 + x + y^3 + 4y - 1 = 0$ has no integer solutions

Size $h = 35$, length $l = 11$

Theorem 1. *The equation*

$$x^4 + x + y^3 + 4y - 1 = 0 \tag{1}$$

has no integer solutions.

The proof uses five polynomial identities of the form $H_k T_k = U_k^2 + d_k V_k^2$ on the curve (1), where $d_1 = 2H_2$, $d_2 = -2H_1$, $d_3 = -H_4$, $d_4 = -19H_3$, and $d_5 = -19$, and the product formula for three Hilbert symbols. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only basic properties of the Legendre symbol and of the Hilbert symbol.

1 Legendre symbols and Hilbert symbols

For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. It is multiplicative in a , and it depends only on a modulo p [1, Chapter 5].

For a prime p , let $\mathbb{Q}_p$ be the field of p -adic numbers, and put $\mathbb{Q}_\infty = \mathbb{R}$. For v a prime or $v = \infty$, and non-zero $a, b \in \mathbb{Q}$, the Hilbert symbol $(a, b)_v$ is 1 if the equation $z^2 = ar^2 + bs^2$ has a solution $(r, s, z) \neq (0, 0, 0)$ in $\mathbb{Q}_v$, and -1 otherwise. In particular, $(a, b)_v = 1$ for every v if this equation has a non-zero solution in $\mathbb{Q}$. For an odd integer t , we put $\varepsilon(t) = (t - 1)/2$ and $\omega(t) = (t^2 - 1)/8$. We use the following facts [2, Chapter III, Proposition 2 and Theorems 1 and 3].

Theorem 2. *Let a, a' and b be non-zero integers, and let v be a prime or $v = \infty$.*

- (i) $(a, b)_\infty = -1$ if and only if $a < 0$ and $b < 0$.
- (ii) Let p be an odd prime, and write $a = p^\alpha u$ and $b = p^\beta w$ with $p \nmid uw$. Then $(a, b)_p = (-1)^{\alpha\beta\varepsilon(p)}(u/p)^\beta(w/p)^\alpha$.
- (iii) Write $a = 2^\alpha u$ and $b = 2^\beta w$ with u and w odd. Then $(a, b)_2 = (-1)^{\varepsilon(u)\varepsilon(w) + \alpha\omega(w) + \beta\omega(u)}$.
- (iv) $(a, b)_v = 1$ for all but finitely many v , and $\prod_v (a, b)_v = 1$, the product over all primes v and $v = \infty$.
- (v) $(a, b)_v = (b, a)_v$ and $(aa', b)_v = (a, b)_v(a', b)_v$.

Lemma 3. *Let p be an odd prime, and let a, b, r, c, t be integers with $a \neq 0$, $b \neq 0$ and $at = r^2 - bc^2$. If $p \nmid b$, and p does not divide both r and c , then $(a, b)_p = 1$.*

Proof. If $p \nmid a$, this is Theorem 2(ii) with $\alpha = \beta = 0$. If $p \mid a$, then $r^2 \equiv bc^2 \pmod{p}$. If $p \mid c$, then $p \mid r$, which is excluded. Hence $p \nmid c$, so $b \equiv (rc')^2 \pmod{p}$, where $cc' \equiv 1 \pmod{p}$, and $(b/p) = 1$ as $p \nmid b$. Theorem 2(ii) with $\beta = 0$ gives $(a, b)_p = (b/p)^{v_p(a)} = 1$. $\square$

Lemma 4. Let a, b, r, c, t be integers with $at = r^2 - bc^2 \neq 0$ and $b \neq 0$. Then $(a, b)_v = (t, b)_v$ for every v .

Proof. The triple $(r, s, z) = (1, c, r)$ is a non-zero rational solution of $z^2 = (at)r^2 + bs^2$, so $(at, b)_v = 1$. By Theorem 2(v), $(a, b)_v(t, b)_v = (at, b)_v = 1$, and both symbols are ± 1 . $\square$

Lemma 5. Let a and b be non-zero integers. If p is an odd prime with $p \nmid a$ and $(a/p) = 1$, then $(a, b)_p = 1$. If $a \equiv 1 \pmod{8}$, then $(a, b)_2 = 1$.

Proof. The first claim is Theorem 2(ii) with $\alpha = 0$. For the second, $\varepsilon(a)$ and $\omega(a)$ are even, and Theorem 2(iii) with $\alpha = 0$ gives $(a, b)_2 = 1$. $\square$

2 The auxiliary polynomials

Put $F(x, y) = x^4 + x + y^3 + 4y - 1$, so that (1) is the equation $F(x, y) = 0$. For the first identity, we define

$$\begin{aligned} H_1 &= -2x - y + 2, \\ T_1 &= 16x^2 - 8xy - 32x + 4y^2 + 8y + 34, \\ U_1 &= 2x^2 - 8x + 8, \\ V_1 &= 1, \\ G_1 &= -4, \end{aligned}$$

for the second identity, we define

$$\begin{aligned} H_2 &= -y, \\ T_2 &= 4y^2 + 14, \\ U_2 &= 2x^2, \\ V_2 &= 1, \\ G_2 &= -4, \end{aligned}$$

for the third identity, we define

$$\begin{aligned} H_3 &= y^2 - y + 2, \\ T_3 &= -y - 1, \\ U_3 &= x^2, \\ V_3 &= 1, \\ G_3 &= -1, \end{aligned}$$

for the fourth identity, we define

$$\begin{aligned} H_4 &= -x - 3y + 3, \\ T_4 &= 27x^8 - 81x^7y - 163x^7 + 243x^6y^2 + 246x^6y + 1338x^6 \\ &\quad - 9x^5y^2 - 360x^5y - 747x^5 + 1053x^4y^2 + 1053x^4y + 3213x^4 \\ &\quad - 6156y - 6156, \\ U_4 &= -27x^5 + x^4 - 63x^3 + 18x^2, \\ V_4 &= -18y - 18, \\ G_4 &= -729x^6 + 27x^5 - 3159x^4 + 6156y + 6156, \end{aligned}$$

$$\begin{aligned}
\alpha_4 &= -147654x^3 + 3616380x^2 - 889056x - 10153422, \\
\beta_4 &= -221481x^4y^2 + 221481x^4y - 1107405x^4 + 5432773x^3y^2 \\
&\quad - 5432773x^3y + 27163865x^3 - 2051283x^2y^2 + 2051283x^2y \\
&\quad - 10256415x^2 - 2154276xy^2 + 2154276xy - 10771380x - 12925656y^2 \\
&\quad + 12925656y - 64628280, \\
\gamma_4 &= -3986658x^4 + 97789914x^3 - 36923094x^2 - 38776968x - 232661808,
\end{aligned}$$

for the fifth identity, we define

$$\begin{aligned}
H_5 &= -8x - 24y + 5, \\
T_5 &= 3456x^2 - 10368xy - 4320x + 31104y^2 + 6480y + 125766, \\
U_5 &= 864x^2 - 16x + 115, \\
V_5 &= 88x + 83, \\
G_5 &= -746496, \\
\alpha_5 &= 242, \\
\beta_5 &= -2376x + 2285, \\
\gamma_5 &= 0,
\end{aligned}$$

for the auxiliary polynomials H_1 and H_2 together, we define

$$\begin{aligned}
\lambda_1 &= x^3 + x^2 - 4xy + x + 2y^2 + 4y + 2, \\
\mu_1 &= -x^3 + 7x^2 - 17x + 14, \\
\nu_1 &= 2,
\end{aligned}$$

for the auxiliary polynomials H_3 and H_4 together, we define

$$\begin{aligned}
\lambda_2 &= -27x^3 - 53x^2 + 317x + 1950, \\
\mu_2 &= 3x^4 - 9x^3y + 6x^3 - 18x^2y - 36x^2 + xy^2 + 107xy - 211x - y^2 + 649y - 5, \\
\nu_2 &= 3x - 3,
\end{aligned}$$

and, for the signs, we also use

$$\begin{aligned}
Q_1 &= 4x^2 - 2xy - 8x + y^2 + 2y + 8, \\
M_1 &= x^4 - 8x^3 + 24x^2 - 31x + 15, \\
E_1 &= 12x^2 - 24x + 28, \\
Q_2 &= y^2 + 4, \\
M_2 &= x^4 + x - 1, \\
E_2 &= 16, \\
Q_4 &= x^2 - 3xy - 6x + 9y^2 + 9y + 45, \\
M_4 &= 27x^4 - x^3 + 9x^2 - 36x + 108, \\
E_4 &= 27x^2 - 162x + 1539, \\
Q_5 &= 64x^2 - 192xy - 80x + 576y^2 + 120y + 2329, \\
M_5 &= 13824x^4 - 512x^3 + 960x^2 - 5208x - 2179, \\
E_5 &= 110592x^2 - 138240x + 5351616.
\end{aligned}$$

Expanding both sides, one checks the identities

$$H_1T_1 - U_1^2 - 2H_2V_1^2 = G_1F, \quad (2)$$

$$H_2T_2 - U_2^2 + 2H_1V_2^2 = G_2F, \quad (3)$$

$$H_3T_3 - U_3^2 + H_4V_3^2 = G_3F, \quad (4)$$

$$H_4T_4 - U_4^2 + 19H_3V_4^2 = G_4F, \quad (5)$$

$$H_5T_5 - U_5^2 + 19V_5^2 = G_5F, \quad (6)$$

$$\alpha_4U_4 + \beta_4V_4 + \gamma_4F = 2^5 \cdot 3^8 \cdot 61 \cdot 109, \quad (7)$$

$$\alpha_5U_5 + \beta_5V_5 + \gamma_5F = 3^5 \cdot 5 \cdot 179, \quad (8)$$

$$\lambda_1H_1 + \mu_1H_2 + \nu_1F = 2, \quad (9)$$

$$\lambda_2H_3 + \mu_2H_4 + \nu_2F = 2^4 \cdot 3^5, \quad (10)$$

$$-F = Q_1H_1 - M_1, \quad (11)$$

$$4Q_1 = (-2x + 2y + 2)^2 + E_1. \quad (12)$$

$$-F = Q_2H_2 - M_2, \quad (13)$$

$$4Q_2 = (2y)^2 + E_2. \quad (14)$$

$$4H_3 = (2y - 1)^2 + 7. \quad (15)$$

$$-27F = Q_4H_4 - M_4, \quad (16)$$

$$36Q_4 = (-3x + 18y + 9)^2 + E_4. \quad (17)$$

$$-13824F = Q_5H_5 - M_5, \quad (18)$$

$$2304Q_5 = (-192x + 1152y + 120)^2 + E_5. \quad (19)$$

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1), so that $F(x, y) = 0$; from now on the polynomials denote their values at (x, y) . Put

$$\begin{aligned} A_1 &= -d_2 = 2H_1, & B_1 &= -d_1 = -2H_2, & T_{A_1} &= 2T_1, & T_{B_1} &= -2T_2, \\ A_2 &= -d_4 = 19H_3, & B_2 &= -d_3 = H_4, & T_{A_2} &= 19T_3, & T_{B_2} &= T_4, \\ A_3 &= H_5, & B_3 &= -d_5 = 19, & T_{A_3} &= T_5. \end{aligned}$$

By (2)–(6), after multiplying by squares of the constants c in $d_k = cH_{k'}$ if necessary, we obtain

$$A_1T_{A_1} = (2U_1)^2 - B_1(2V_1)^2, \quad B_1T_{B_1} = (2U_2)^2 - A_1(2V_2)^2. \quad (20)$$

$$A_2T_{A_2} = (19U_3)^2 - B_2(19V_3)^2, \quad B_2T_{B_2} = U_4^2 - A_2V_4^2. \quad (21)$$

$$A_3T_{A_3} = U_5^2 - B_3V_5^2. \quad (22)$$

Step 1: signs. *The sign of H_1 .* Since $F(x, y) = 0$, (11) gives $Q_1H_1 = M_1$, and M_1 and E_1 depend only on x . We show that $M_1 > 0$ and $E_1 > 0$. If $x \geq 2$, write $x = 2 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_1 and E_1 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_1 > 0$ and $E_1 > 0$. It remains to consider $0 \leq x \leq 1$. For $x = 0$ and $x = 1$ we have $M_1 > 0$ and $E_1 > 0$. Hence $M_1 > 0$ and $E_1 > 0$, so (12) gives $4Q_1 \geq E_1 > 0$, and $Q_1H_1 = M_1 > 0$ gives $H_1 > 0$.

The sign of H_2 . Since $F(x, y) = 0$, (13) gives $Q_2H_2 = M_2$, and M_2 and E_2 depend only on x . We show that $M_2 > 0$ and $E_2 > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -2$, write $x = -2 - z$ with $z \geq 0$. In both cases, expanding M_2 and E_2 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_2 > 0$ and $E_2 > 0$. It remains to

consider $-1 \leq x \leq 0$. For $x = -1$ and $x = 0$, we have $F(x, y) \not\equiv 0 \pmod{7}$ for every y , as one checks for the 7 residues of y modulo 7, so (1) has no solution with such x . Hence $M_2 > 0$ and $E_2 > 0$, so (14) gives $4Q_2 \geq E_2 > 0$, and $Q_2H_2 = M_2 > 0$ gives $H_2 > 0$.

The sign of H_3 . By (15), $H_3 > 0$.

The sign of H_4 . Since $F(x, y) = 0$, (16) gives $Q_4H_4 = M_4$, and M_4 and E_4 depend only on x . We show that $M_4 > 0$ and $E_4 > 0$. If $x \geq 3$, write $x = 3 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_4 and E_4 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_4 > 0$ and $E_4 > 0$. It remains to consider $0 \leq x \leq 2$. For $x \in \{0, 1, 2\}$ we have $M_4 > 0$ and $E_4 > 0$. Hence $M_4 > 0$ and $E_4 > 0$, so (17) gives $36Q_4 \geq E_4 > 0$, and $Q_4H_4 = M_4 > 0$ gives $H_4 > 0$.

The sign of H_5 . Since $F(x, y) = 0$, (18) gives $Q_5H_5 = M_5$, and M_5 and E_5 depend only on x . We show that $M_5 > 0$ and $E_5 > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_5 and E_5 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_5 > 0$ and $E_5 > 0$. It remains to consider $x = 0$. For $x = 0$, we have $F(x, y) \not\equiv 0 \pmod{7}$ for every y , as one checks for the 7 residues of y modulo 7, so (1) has no solution with this x . Hence $M_5 > 0$ and $E_5 > 0$, so (19) gives $2304Q_5 \geq E_5 > 0$, and $Q_5H_5 = M_5 > 0$ gives $H_5 > 0$.

By these signs, $A_1 > 0$, $B_1 < 0$, $A_2 > 0$, $B_2 > 0$, and $A_3 > 0$.

Hence Theorem 2(i) gives $(A_1, B_1)_\infty = 1$ (as $A_1 > 0$), $(A_2, B_2)_\infty = 1$ (as $A_2 > 0$), and $(A_3, B_3)_\infty = 1$ (as $A_3 > 0$).

Step 2: primes outside S . Let $S = \{2, 3, 5, 19, 61, 109, 179\}$ be the set of primes dividing 2, the constants 1, 2, and 19 in the $d_k = cH_k$, $d_5 = -19$, $V_1 = 1$, $V_2 = 1$, $V_3 = 1$, the right-hand side of (7), and the right-hand side of (8), and the right-hand sides of (9) and (10). Let $p \notin S$ be an odd prime and $j \in \{1, 2, 3\}$; we show that $(A_j, B_j)_p = 1$. If $p \nmid A_j B_j$, this follows from Theorem 2(ii) with $\alpha = \beta = 0$. For $j = 1$: If $p \mid A_1$, then $p \mid H_1$, so $p \nmid H_2$ by (9) and $p \nmid B_1$; as $p \nmid 2V_1$, Lemma 3 with $(a, b, r, c, t) = (A_1, B_1, 2U_1, 2V_1, T_{A_1})$ and (20) gives $(A_1, B_1)_p = 1$; if $p \mid B_1$, then $p \mid H_2$, so $p \nmid H_1$ by (9) and $p \nmid A_1$; as $p \nmid 2V_2$, Lemma 3 with $(a, b, r, c, t) = (B_1, A_1, 2U_2, 2V_2, T_{B_1})$ and (20) gives $(B_1, A_1)_p = 1$, which equals $(A_1, B_1)_p$ by Theorem 2(v). For $j = 2$: If $p \mid A_2$, then $p \mid H_3$, so $p \nmid H_4$ by (10) and $p \nmid B_2$; as $p \nmid 19V_3$, Lemma 3 with $(a, b, r, c, t) = (A_2, B_2, 19U_3, 19V_3, T_{A_2})$ and (21) gives $(A_2, B_2)_p = 1$; if $p \mid B_2$, then $p \mid H_4$, so $p \nmid H_3$ by (10) and $p \nmid A_2$; as p does not divide both U_4 and V_4 by (7), Lemma 3 with $(a, b, r, c, t) = (B_2, A_2, U_4, V_4, T_{B_2})$ and (21) gives $(B_2, A_2)_p = 1$, which equals $(A_2, B_2)_p$ by Theorem 2(v). For $j = 3$: Here $p \nmid B_3$ as $p \notin S$. If $p \mid A_3$, then, as p does not divide both U_5 and V_5 by (8), Lemma 3 with $(a, b, r, c, t) = (A_3, B_3, U_5, V_5, T_{A_3})$ and (22) gives $(A_3, B_3)_p = 1$.

Step 3: the primes in S . By Step 1, the A_j and B_j are non-zero. If $T_{A_j} \neq 0$, then (20)–(22) and Lemma 4 give $(A_j, B_j)_v = (T_{A_j}, B_j)_v$, and if $T_{B_j} \neq 0$, they give $(A_j, B_j)_v = (B_j, A_j)_v = (T_{B_j}, A_j)_v = (A_j, T_{B_j})_v$, using Theorem 2(v). For a non-zero integer X and a prime p , we write $X \in p^{2i}(\pm)$ or $X \in p^{2i+1}(\pm)$, when p is odd, if $X = p^j u$ with $p \nmid u$, $(u/p) = \pm 1$, and j even or odd, respectively; and we write $X \in 2^{2i} \cdot u$ or $X \in 2^{2i+1} \cdot u$ if $X = 2^j u'$ with $u' \equiv u \pmod{8}$ and j even or odd. If $X \not\equiv 0 \pmod{p^e}$ and $v_p(X) \leq e - 1$, or $v_p(X) \leq e - 3$ if $p = 2$, then this class is determined by X modulo p^e . By Theorem 2(ii) and (iii), the classes of X and Y determine $(X, Y)_p$.

The prime 2. Checking the 128^2 residue pairs modulo 128, we find that (1) has exactly 128 solutions modulo 128, and that at each of them, for each j , one of the cases listed for $p = 2$ and j in Table 1 holds; moreover, the combination of these cases is one of the combinations listed there. Each case gives the stated value of $(A_j, B_j)_2$. Hence $\sigma_2 \in V_2$, with V_2 as in Step 4.

The prime 3. Checking the 3^2 residue pairs modulo 3, we find that (1) has exactly 3 solutions modulo 3, and that at each of them, for each j , one of the cases listed for $p = 3$ and j in Table 1 holds. Each case gives the stated value of $(A_j, B_j)_3$. Hence $\sigma_3 \in V_3$, with V_3 as in Step 4.

The prime 5. Checking the 5^2 residue pairs modulo 5, we find that (1) has exactly 3 solutions modulo 5, and that at each of them, for each j , one of the cases listed for $p = 5$ and j in Table 1 holds. Each case gives the stated value of $(A_j, B_j)_5$. Hence $\sigma_5 \in V_5$, with V_5 as in Step 4.

The prime 19. Checking the 361^2 residue pairs modulo 361, we find that (1) has exactly 342 solutions modulo 361, and that at each of them, for each j , one of the cases listed for $p = 19$ and j in Table 1 holds; moreover, the combination of these cases is one of the combinations listed there. Each case gives the stated value of $(A_j, B_j)_{19}$. Hence $\sigma_{19} \in V_{19}$, with V_{19} as in Step 4.

The prime 61. Checking the 61^2 residue pairs modulo 61, we find that (1) has exactly 63 solutions modulo 61, and that at each of them, for each j , one of the cases listed for $p = 61$ and j in Table 1 holds. Each case gives the stated value of $(A_j, B_j)_{61}$. Hence $\sigma_{61} \in V_{61}$, with V_{61} as in Step 4.

The prime 109. Checking the 109^2 residue pairs modulo 109, we find that (1) has exactly 94 solutions modulo 109, and that at each of them, for each j , one of the cases listed for $p = 109$ and j in Table 1 holds. Each case gives the stated value of $(A_j, B_j)_{109}$. Hence $\sigma_{109} \in V_{109}$, with V_{109} as in Step 4.

The prime 179. Checking the 179^2 residue pairs modulo 179, we find that (1) has exactly 196 solutions modulo 179, and that at each of them, for each j , one of the cases listed for $p = 179$ and j in Table 1 holds. Each case gives the stated value of $(A_j, B_j)_{179}$. Hence $\sigma_{179} \in V_{179}$, with V_{179} as in Step 4.

Step 4: contradiction. For $p \in S$ and $p = \infty$, put

$$\sigma_p = ((A_1, B_1)_p, (A_2, B_2)_p, (A_3, B_3)_p).$$

By Theorem 2(iv), applied to each j , and by Step 2, the componentwise product of the σ_p with $p \in S$ is σ_∞ , which is $(1, 1, 1)$ by Step 1. By Step 3, $\sigma_p \in V_p$, where

$$\begin{aligned} V_2 &= \{(1, 1, 1), (-1, 1, 1), \\ &\quad (-1, -1, 1)\}, \\ V_3 &= \{(1, 1, 1)\}, \\ V_5 &= \{(1, 1, 1)\}, \\ V_{19} &= \{(1, 1, -1), (1, -1, 1)\}, \\ V_{61} &= \{(1, 1, 1)\}, \\ V_{109} &= \{(1, 1, 1)\}, \\ V_{179} &= \{(1, 1, 1)\}. \end{aligned}$$

The componentwise products of one vector from each of V_3 , V_5 , V_{19} , V_{61} , V_{109} , and V_{179} are $(1, 1, -1)$ and $(1, -1, 1)$, so σ_2 would have to be one of $(1, 1, -1)$ and $(1, -1, 1)$, but none of them is in V_2 . This contradiction proves the theorem. $\square$

References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.
- [2] J.-P. Serre, *A Course in Arithmetic*, Graduate Texts in Mathematics 7, Springer, New York, 1973.

Table 1: The cases in Step 3. In each case, the symbol $(A_j, B_j)_p$ is computed as the stated symbol, whose entries lie in the stated classes; an entry “any” means that the other entry is a unit square, so that Lemma 5 applies.

	symbol	first entry	second entry	value
$p = 2$, solutions modulo 128				
$j = 1$				
1	(A_1, B_1)	$2^{2i+1} \cdot 1$	$2^{2i+1} \cdot 1$	1
2	(A_1, B_1)	$2^{2i+1} \cdot 1$	$2^{2i+1} \cdot 7$	1
3	(A_1, B_1)	$2^{2i+1} \cdot 3$	$2^{2i+1} \cdot 1$	-1
4	(A_1, B_1)	$2^{2i+1} \cdot 3$	$2^{2i+1} \cdot 3$	-1
5	(A_1, B_1)	$2^{2i+1} \cdot 5$	$2^{2i+1} \cdot 3$	1
6	(A_1, B_1)	$2^{2i+1} \cdot 5$	$2^{2i+1} \cdot 5$	1
7	(A_1, B_1)	$2^{2i+1} \cdot 7$	$2^{2i+1} \cdot 5$	-1
8	(A_1, B_1)	$2^{2i+1} \cdot 7$	$2^{2i+1} \cdot 7$	-1
$j = 2$				
1	(A_2, B_2)	$2^{2i} \cdot 1$	$2^{2i} \cdot 1$	1
2	(A_2, B_2)	$2^{2i} \cdot 1$	$2^{2i} \cdot 5$	1
3	(A_2, B_2)	$2^{2i} \cdot 1$	$2^{2i} \cdot 7$	1
4	(A_2, B_2)	$2^{2i} \cdot 3$	$2^{2i} \cdot 1$	1
5	(A_2, B_2)	$2^{2i} \cdot 3$	$2^{2i} \cdot 5$	1
6	(A_2, B_2)	$2^{2i} \cdot 5$	$2^{2i} \cdot 1$	1
7	(A_2, B_2)	$2^{2i} \cdot 5$	$2^{2i} \cdot 3$	1
8	(A_2, B_2)	$2^{2i} \cdot 5$	$2^{2i} \cdot 5$	1
9	(A_2, B_2)	$2^{2i} \cdot 7$	$2^{2i} \cdot 1$	1
10	(A_2, B_2)	$2^{2i} \cdot 7$	$2^{2i} \cdot 5$	1
11	(A_2, B_2)	$2^{2i+1} \cdot 1$	$2^{2i} \cdot 1$	1
12	(A_2, B_2)	$2^{2i+1} \cdot 1$	$2^{2i} \cdot 5$	-1
13	(A_2, B_2)	$2^{2i+1} \cdot 1$	$2^{2i} \cdot 7$	1
14	(A_2, B_2)	$2^{2i+1} \cdot 3$	$2^{2i} \cdot 1$	1
15	(A_2, B_2)	$2^{2i+1} \cdot 3$	$2^{2i} \cdot 3$	1
16	(A_2, B_2)	$2^{2i+1} \cdot 3$	$2^{2i} \cdot 5$	-1
17	(A_2, B_2)	$2^{2i+1} \cdot 5$	$2^{2i} \cdot 1$	1
18	(A_2, B_2)	$2^{2i+1} \cdot 5$	$2^{2i} \cdot 5$	-1
19	(A_2, B_2)	$2^{2i+1} \cdot 5$	$2^{2i} \cdot 7$	1
20	(A_2, B_2)	$2^{2i+1} \cdot 7$	$2^{2i} \cdot 1$	1
21	(A_2, B_2)	$2^{2i+1} \cdot 7$	$2^{2i} \cdot 3$	1
22	(A_2, B_2)	$2^{2i+1} \cdot 7$	$2^{2i} \cdot 5$	-1
23	(T_{A_2}, B_2)	$2^{2i} \cdot 3$	$2^{2i} \cdot 1$	1
$j = 3$				
1	(A_3, B_3)	$2^{2i} \cdot 5$	$2^{2i} \cdot 3$	1
combinations of cases ($j = 1, j = 2, j = 3$) and the vector σ_2				
	$(1, 14, 1) \mapsto (1, 1, 1)$	$(1, 20, 1) \mapsto (1, 1, 1)$		
	$(2, 1, 1) \mapsto (1, 1, 1)$	$(2, 4, 1) \mapsto (1, 1, 1)$		
	$(2, 6, 1) \mapsto (1, 1, 1)$	$(2, 9, 1) \mapsto (1, 1, 1)$		
	$(3, 16, 1) \mapsto (-1, -1, 1)$	$(3, 22, 1) \mapsto (-1, -1, 1)$		
	$(4, 2, 1) \mapsto (-1, 1, 1)$	$(4, 5, 1) \mapsto (-1, 1, 1)$		
	$(4, 8, 1) \mapsto (-1, 1, 1)$	$(4, 10, 1) \mapsto (-1, 1, 1)$		
	$(4, 13, 1) \mapsto (-1, 1, 1)$	$(4, 15, 1) \mapsto (-1, 1, 1)$		
	$(4, 19, 1) \mapsto (-1, 1, 1)$	$(4, 21, 1) \mapsto (-1, 1, 1)$		
	$(4, 23, 1) \mapsto (-1, 1, 1)$	$(5, 1, 1) \mapsto (1, 1, 1)$		
	$(5, 4, 1) \mapsto (1, 1, 1)$	$(5, 6, 1) \mapsto (1, 1, 1)$		
	$(5, 9, 1) \mapsto (1, 1, 1)$	$(5, 11, 1) \mapsto (1, 1, 1)$		
	$(5, 14, 1) \mapsto (1, 1, 1)$	$(5, 17, 1) \mapsto (1, 1, 1)$		
	$(5, 20, 1) \mapsto (1, 1, 1)$	$(5, 23, 1) \mapsto (1, 1, 1)$		
	$(6, 11, 1) \mapsto (1, 1, 1)$	$(6, 17, 1) \mapsto (1, 1, 1)$		
	$(7, 12, 1) \mapsto (-1, -1, 1)$	$(7, 18, 1) \mapsto (-1, -1, 1)$		
	$(8, 3, 1) \mapsto (-1, 1, 1)$	$(8, 4, 1) \mapsto (-1, 1, 1)$		

symbol	first entry	second entry	value
$(8, 7, 1) \mapsto (-1, 1, 1)$		$(8, 10, 1) \mapsto (-1, 1, 1)$	
$p = 3$, solutions modulo 3			
$j = 1$			
1 $3 \nmid A_1 B_1$			1
2 (A_1, B_1)	any	$3^{2i}(+)$	1
$j = 2$			
1 $3 \nmid A_2 B_2$			1
2 (A_2, B_2)	$3^{2i}(+)$	any	1
$j = 3$			
1 $3 \nmid A_3 B_3$			1
2 (A_3, B_3)	any	$3^{2i}(+)$	1
$p = 5$, solutions modulo 5			
$j = 1$			
1 $5 \nmid A_1 B_1$			1
2 (T_{A_1}, B_1)	$5^{2i}(-)$	$5^{2i}(+)$	1
$j = 2$			
1 $5 \nmid A_2 B_2$			1
$j = 3$			
1 $5 \nmid A_3 B_3$			1
2 (T_{A_3}, B_3)	$5^{2i}(+)$	$5^{2i}(+)$	1
$p = 19$, solutions modulo 361			
$j = 1$			
1 $19 \nmid A_1 B_1$			1
2 (A_1, B_1)	$19^{2i+1}(-)$	$19^{2i}(+)$	1
3 (A_1, B_1)	$19^{2i+1}(+)$	$19^{2i}(+)$	1
4 (T_{A_1}, B_1)	$19^{2i}(+)$	$19^{2i}(+)$	1
$j = 2$			
1 (A_2, B_2)	$19^{2i+1}(-)$	$19^{2i}(-)$	-1
2 (A_2, B_2)	$19^{2i+1}(-)$	$19^{2i}(+)$	1
3 (A_2, B_2)	$19^{2i+1}(-)$	$19^{2i+1}(+)$	1
4 (A_2, B_2)	$19^{2i+1}(+)$	$19^{2i}(-)$	-1
5 (A_2, B_2)	$19^{2i+1}(+)$	$19^{2i}(+)$	1
$j = 3$			
1 (A_3, B_3)	$19^{2i}(-)$	$19^{2i+1}(+)$	-1
2 (A_3, B_3)	$19^{2i}(+)$	$19^{2i+1}(+)$	1
3 (A_3, B_3)	$19^{2i+1}(+)$	$19^{2i+1}(+)$	-1
combinations of cases ($j = 1, j = 2, j = 3$) and the vector σ_{19}			
$(1, 1, 2) \mapsto (1, -1, 1)$		$(1, 2, 1) \mapsto (1, 1, -1)$	
$(1, 3, 3) \mapsto (1, 1, -1)$		$(1, 4, 2) \mapsto (1, -1, 1)$	
$(1, 5, 1) \mapsto (1, 1, -1)$		$(2, 5, 1) \mapsto (1, 1, -1)$	
$(3, 5, 1) \mapsto (1, 1, -1)$		$(4, 5, 1) \mapsto (1, 1, -1)$	
$p = 61$, solutions modulo 61			
$j = 1$			
1 $61 \nmid A_1 B_1$			1
2 (A_1, T_{B_1})	$61^{2i}(+)$	$61^{2i}(-)$	1
3 (T_{A_1}, B_1)	$61^{2i}(+)$	$61^{2i}(+)$	1
$j = 2$			
1 $61 \nmid A_2 B_2$			1
$j = 3$			
1 $61 \nmid A_3 B_3$			1
2 (T_{A_3}, B_3)	$61^{2i}(-)$	$61^{2i}(+)$	1
3 (T_{A_3}, B_3)	$61^{2i}(+)$	$61^{2i}(+)$	1
$p = 109$, solutions modulo 109			
$j = 1$			
1 $109 \nmid A_1 B_1$			1
2 (T_{A_1}, B_1)	$109^{2i}(+)$	$109^{2i}(+)$	1

	symbol	first entry	second entry	value
$j = 2$				
1	$109 \nmid A_2 B_2$			1
2	(A_2, T_{B_2})	$109^{2i}(+)$	$109^{2i}(-)$	1
3	(T_{A_2}, B_2)	$109^{2i}(-)$	$109^{2i}(+)$	1
$j = 3$				
1	$109 \nmid A_3 B_3$			1
$p = 179$, solutions modulo 179				
$j = 1$				
1	$179 \nmid A_1 B_1$			1
2	(A_1, T_{B_1})	$179^{2i}(+)$	$179^{2i}(+)$	1
$j = 2$				
1	$179 \nmid A_2 B_2$			1
2	(T_{A_2}, B_2)	$179^{2i}(+)$	$179^{2i}(+)$	1
$j = 3$				
1	$179 \nmid A_3 B_3$			1
2	(T_{A_3}, B_3)	$179^{2i}(-)$	$179^{2i}(+)$	1
3	(T_{A_3}, B_3)	$179^{2i}(+)$	$179^{2i}(+)$	1